

ME325: Mechanical Engineering Design Project

Design of Efficient Air Mixing & Supply System for Tea Withering

St. Coomb Tea Factory, TRI Thalawakelle



E/21/086 | E/21/215 | E/21/265

PROBLEM STATEMENT & PROJECT AIM

The Need: Current manual withering systems result in **30% energy loss** and non-uniform moisture removal due to poor hot/ambient air blending.

Operational Feature	Existing Manual System	Proposed Engineered Solution
Air Mixing	Inconsistent / Uncontrolled	Automated Mixing Chamber
Leaf Agitation	Manual Turning (High Labor)	Mechanical Turning Mechanism
Uniformity	Temperature Variation (Hot Spots)	Uniform Distribution via Redesigned Ducts
Monitoring	Subjective / Human Judgment	Sensor-Based Real-time Control

ENGINEERING DESIGN

Our methodology focuses on maximizing **80% visual analysis** through Computational Fluid Dynamics (CFD). We integrate a controlled mixing chamber with optimized dampers and fan placement.

Primary Software Stack

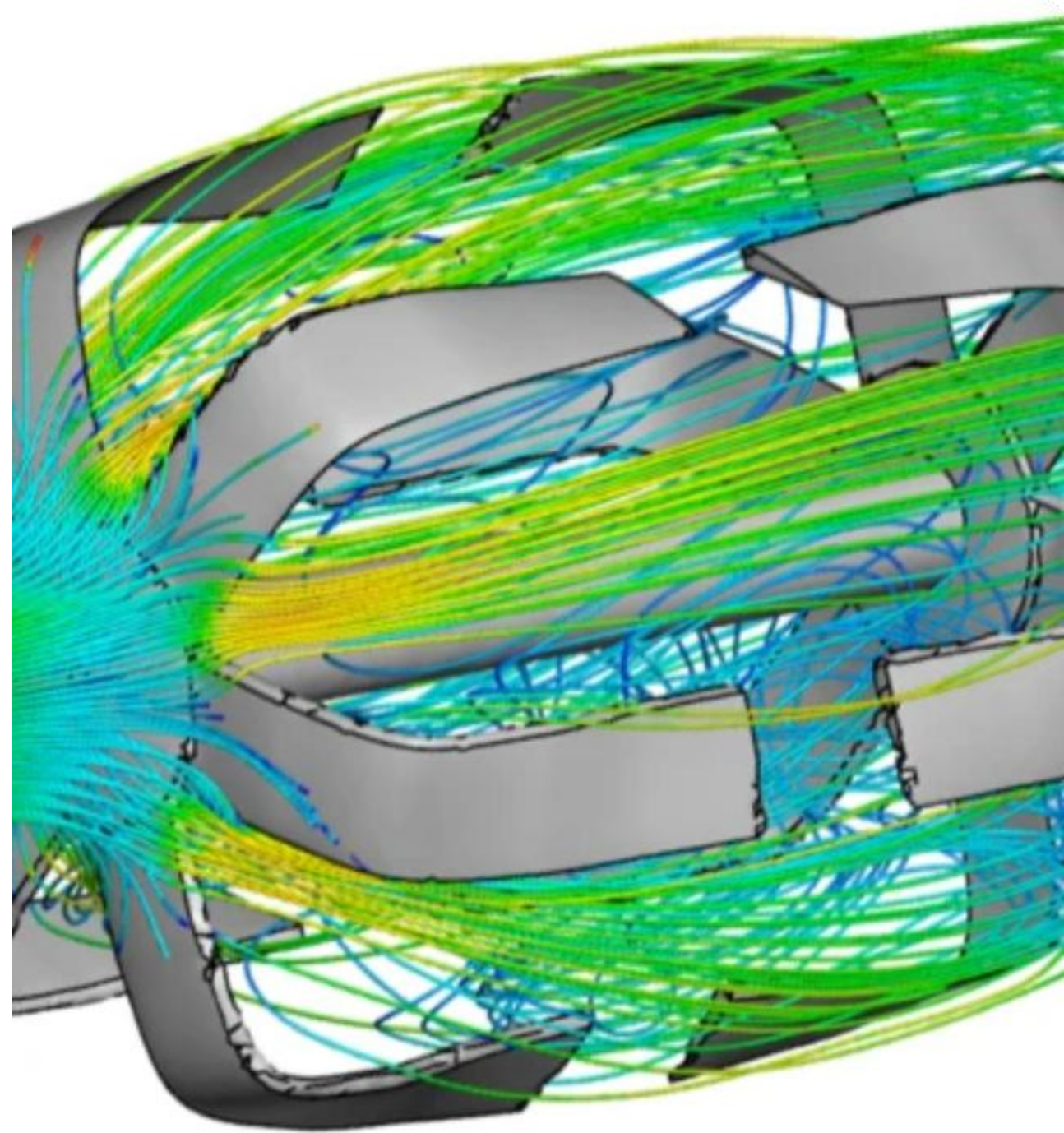
SolidWorks

ANSYS CFD

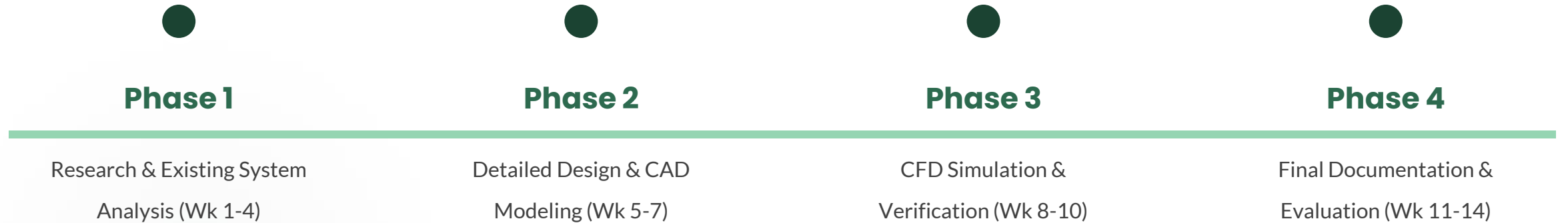
MATLAB

Fundamental Airflow Volume Flow Rate:

$$Q = A \cdot v$$



CRITICAL PATH & PROJECT TIMELINE



IMPACT ANALYSIS: THE "BIG THREE"



Economic

Significant reduction in labor costs and energy wastage. Optimized withering time increases daily factory throughput and market viability.



Social

Elimination of strenuous manual leaf turning improves workplace ergonomics and operator safety, reducing long-term occupational fatigue.



Environmental

Enhanced heat distribution ensures sustainable energy utilization. Improved moisture control reduces waste and supports eco-friendly manufacturing.